

Database Technology Research

1. PostgreSQL (Relational Database)

PostgreSQL, the highly developed open-source relational database, is celebrated for its reliability and strong data consistency.

Project use cases:

- User profiles and accounts
- Questions, activities, and lessons
- Attempts, scores, and progress monitoring

Benefits:

- Conformity to ACID principles
- Powerful support for relational data
- Sophisticated indexing for queries in analytics

Due to the need for structured and consistent data in learning progress and assessments, PostgreSQL was chosen. (Stonebraker and Kemnitz, 1991).

2. NoSQL Databases (MongoDB / Firebase)

The unstructured data and large data such as analytics events were the main reasons for the investigation of NoSQL databases for managing purposes.

Cases where the project can be applied:

- Users' activity logs
- Voice interaction events with virtual assistants
- Checking user retention

Benefits:

- Unmatched scalability
- Changeable structure
- Fast writing speed

Through the use of polyglot persistence method, Firebase and MongoDB were considered as a backup database for PostgreSQL. (Sadalage and Fowler, 2012).